Cross-Level Interaction and Intralevel Fusion Network for Remote Sensing Image Dehazing

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***Abstract*— Existing approaches have significantly advanced remote sensing image dehazing. However, they often rely on conventional encoder–decoder architectures, leading to prolonged inference times. To this end, we propose a novel cross-level interaction and intralevel fusion network (CINet) for remote sensing image dehazing, which shifts the focus from encoder–decoder dependencies to an innovative hierarchical architecture centered on skip connections, leading to competitive dehazing performance with decreased calculating complexity. Furthermore, we introduce a cross-level multiscale interaction module (CMIM) to facilitate effective interactions between features across hierarchical levels, mitigating the information loss commonly caused by repeated downsampling operations. Meanwhile, we develop an intralevel dual-dimension fusion module (IDFM), which leverages height-wise and width-wise self-attention to capture rich spatial-aware information, enabling robust and efficient intralevel feature fusion. Additionally, we propose a multiview progressive extraction block (MPEB), which decomposes features into four distinct components and applies convolutions with diverse kernel sizes, groups, and dilation factors. This design promotes progressive feature learning while significantly reducing computational overhead. Extensive experiments conducted on nine publicly available datasets validate the effectiveness and superiority of our proposed model.**

***Index Terms*— Cross-level interaction, image dehazing, intralevel fusion, multiview learning, remote sensing image.**

I. Introduction

REMOTE sensing technology is of paramount significance across numerous domains [1], [2], [3], including meteorological and climate forecasting, agricultural productivity, monitoring of natural disasters, exploration of forest and mineral resources, as well as military reconnaissance.

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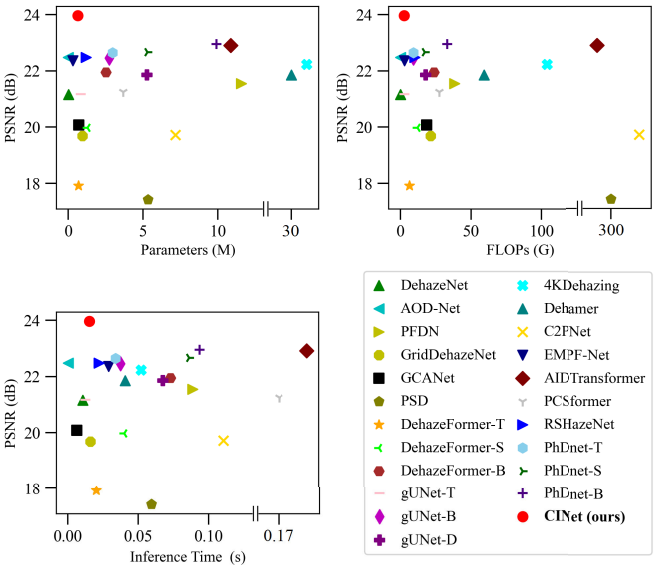


Fig. 1. Comparative analysis of the efficiency of remote sensing image dehazing methods reveals that our approach outperforms others in both metric scores and computational efficiency.

However, remote sensing images are susceptible to diverse factors, primarily stemming from the extended light path, notably atmospheric conditions and climate-related deteriorations. In the field of remote sensing image enhancement, haze is a common meteorological phenomenon that substantially degrades the image quality. It diminishes the image contrast and color fidelity, thereby hindering critical vision tasks such as image classification [4], [5], [6], [7], modality translation [8], [9], and image super-resolution [10], [11], [12]. Therefore, remote sensing image dehazing plays a vital role in improving the quality of remote sensing images captured under hazy conditions.

Over the past few years, a plenty of dehazing strategies have emerged, which can be broadly classified into two categories, model-based and learning-based methods. The former approaches primarily revolve around the atmospheric scattering model [13], [14], [15], [16], which can be defined as

$$\boldsymbol{I}(x) = \boldsymbol{J}(x)\boldsymbol{t}(x) + A(1 - \boldsymbol{t}(x)) \tag{1}$$

where x represents the pixel location, while $\boldsymbol{I}(x)$ and $\boldsymbol{J}(x)$ denote the haze-degraded and haze-free images, respectively.

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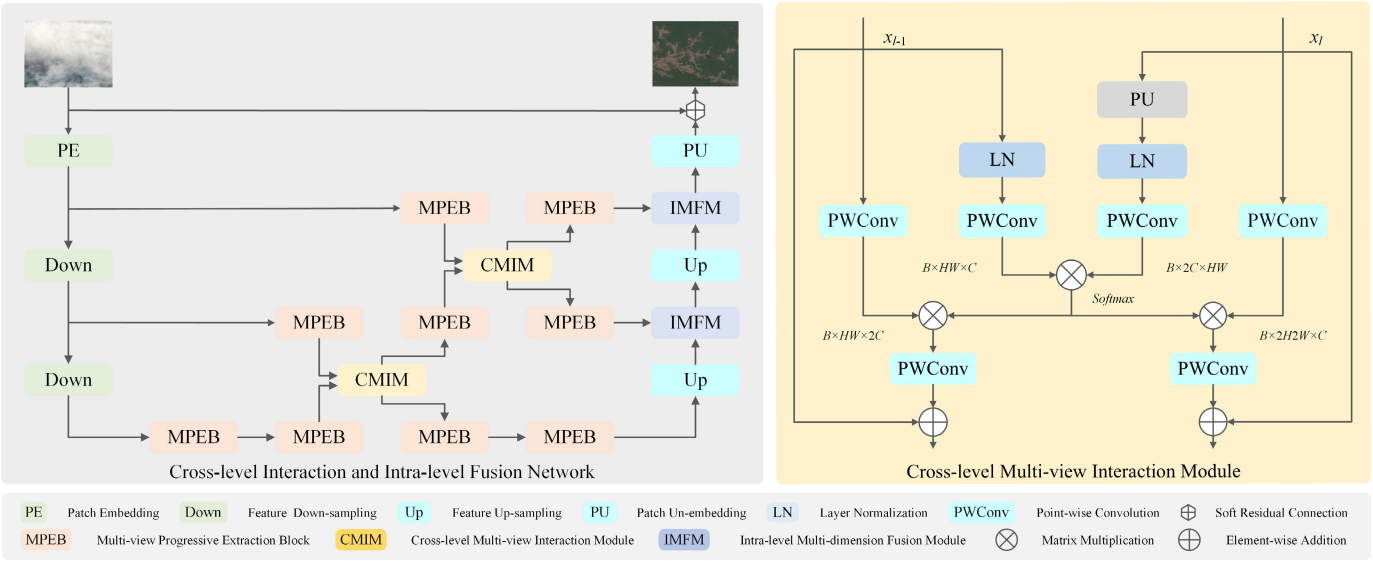


Fig. 2. Architectural overview of the proposed CINet for remote sensing image dehazing. Our approach emphasizes the development of optimal skip connections to regulate the downsampling information flowing into the upsampling stage, which comprises three novel modules, MPEB, IDFM, and CMIM.

The variable A represents the global atmospheric light, and $\boldsymbol{t}(x)$ signifies the medium transmission map. The formulation of $\boldsymbol{t}(x)$ can be formulated as

$$\boldsymbol{t}(x) = e^{-\beta \boldsymbol{d}(x)} \tag{2}$$

where the scattering coefficient of atmosphere is denoted by β, while $\boldsymbol{d}(x)$ represents the scene depth. Unfortunately, the estimation of accurate parameters (#Param) poses challenges, leading to unsatisfactory outcomes in model-based methods [13], [17], [18], [19], [20], particularly when confronted with complicated remote sensing hazy conditions.

Recently, significant progress has been made in learning-based approaches for image dehazing [16], [21], [22], [23], [24], [25]. Among these advancements, DehazeNet [26] leverages convolutional neural networks to estimate the transmission map from hazy images. In contrast, AOD-Net reformulates the image degradation model (1) to enable the network to estimate a single component, facilitating an end-to-end dehazing process. Furthermore, GridDehazeNet [27] challenges the conventional approach by proposing that directly restoring the image is more effective than estimating $\boldsymbol{t}(x)$, as the latter can lead to suboptimal results. Compared to natural hazy images, remote sensing images typically feature higher resolutions and exhibit greater variability and nonuniformity in haze intensity and distribution. Therefore, most existing dehazing methods are not well-suited for the unique challenges posed by remote sensing image dehazing. To this end, DADN [28] introduces a prior-driven dense attentive network that combines dense blocks and attention mechanisms to recover haze-free images. Meanwhile, FCTF-Net [29] adopts a two-stage framework, while DCRD-Net [30] employs a dual-step cascaded residual dense network for direct background reconstruction. Furthermore, H2RL-Net [31] utilizes a multiscale architecture to alleviate haze degradation in remote sensing images, and M2SCN [32] integrates a multimodel joint estimation module and a self-correcting module to develop an end-to-end dehazing network. More recently, DehazeFormer [2] introduces a novel normalization layer and a spatial information aggregation strategy, achieving the state-of-the-art performance for both natural and remote sensing image dehazing tasks. Despite these advancements, existing methods continue to struggle with achieving an optimal trade-off between inference efficiency and image restoration quality, particularly when processing high-resolution remote sensing hazy images.

In this work, we challenge the prevailing u-shaped design strategies [2], [33], [34], [35], which heavily emphasize encoders and decoders, by shifting our focus to skip connections as a mechanism for regulating information flow. Our encoder-minimal and decoder-minimal approach simplifies the architecture by reducing the complexity of managing block quantities during encoding and decoding stages, which achieves the best trade-off between the image quality and model efficiency (see Fig. 1). As shown in Fig. 2, our design streamlines the encoding and decoding processes to consist solely of downsampling and upsampling operations, eliminating unnecessary computational overhead. Additionally, we address the computational inefficiencies of Transformer-based methods [2], which hinder the practical deployment of remote sensing image dehazing techniques. While recent work [36] introduces partial convolution to transform only one quarter of feature planes, leaving the rest unchanged for subsequent layers, we argue that this feature constancy approach is suboptimal for remote sensing image dehazing. Instead, we emphasize the importance of multiview features for accurate information learning in this domain. To overcome these limitations, we propose a novel multiview progressive extraction block (MPEB). Unlike conventional methods that rely on computationally expensive 3×3 convolutions, our MPEB leverages lightweight 1×1 convolutions for efficient channel-wise interactions. Furthermore, to capture multiview features from the remaining three-quarters of input features, we employ grouped convolutions with diverse receptive fields and dilation rates. This design enables comprehensive and

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accurate feature extraction from multiple perspectives, enhancing the ability of basic learning block to represent the input features robustly. By integrating MPEB, our approach not only replicates the channel-wise interaction capabilities of single-view convolution methods like [36] but also introduces the ability to extract multiview features. This dual capability allows the network to represent input features from multiple perspectives, resulting in more robust and discriminative feature representations.

In conventional u-shaped architectures, standard methods typically fuse intralevel features by concatenating them along the channel dimension, followed by a 1×1 convolution to conduct feature interaction from skip connections and upsampling operations. However, this approach struggles to effectively integrate information across different feature planes and fail to capture context-aware attention, limiting the model ability to fully utilize spatial and contextual relationships. To address these challenges, we introduce an intralevel dual-dimension fusion module (IDFM), which conducts feature fusion with both height-wise and width-wise self-attention. This design not only significantly reduces the computational complexity of self-attention but also ensures that more accurate and structure-preserving features are propagated to the decoding stages. As a result, our proposed IDFM enables more robust and precise intralevel feature integration, ultimately leading to significant improvements in remote sensing image dehazing performance.

In the field of remote sensing image dehazing, many existing methods [2], [32], [37] struggle to establish effective interactions across different hierarchical levels, which limits the flow of information among multiscale features. This limitation significantly hinders network performance optimization, as the seamless exchange of information is disrupted. To overcome this challenge, we introduce the cross-level multiscale interaction module (CMIM) to enhance the feature interactions between features with different resolutions. Specifically, CMIM leverages the transposed self-attention to compute attention maps, providing a more computationally efficient approach compared to traditional spatial self-attention [16], [38]. In this module, the query representation ($\boldsymbol{Q}$) is derived from input features at level l, and the key representation ($\boldsymbol{K}$) is obtained from level $l-1$, while two value representations ($\boldsymbol{V}$) are enhanced using the same mutual attention map, ensuring robust information exchanges across different hierarchical stages. This design effectively mitigates the information loss caused by repetitive sampling operations, producing two output features that maintain the resolutions of their original inputs. By enabling efficient and seamless information flow, CMIM significantly enhances the model ability to process and optimize multiscale features in remote sensing image dehazing.

This work is an extension of our previous work [39]. The overall architecture of our proposed cross-level interaction and intralevel fusion network (CINet) for remote sensing image dehazing is depicted in Fig. 2. The main contributions of our work can be summarized as follows.

1) We propose an innovative architectural design that prioritizes the skip connections, which enhances the regulation of information flow between the learning and reconstruction stages.
2) We present a multiview progressive extraction module that performs feature extraction across multiple perspectives.
3) An IDFM designed to integrate global information, thereby enhancing the fusion process with greater comprehensiveness and context-awareness.
4) A cross-level multiscale interaction module that facilitates effective communication and collaboration across different levels, ensuring the retention and sharing of crucial information.

II. Related Work

A. Model-Based Remote Sensing Image Dehazing

Model-based approaches predominantly rely on the estimation of the transmission map, encompassing prominent techniques such as dark channel priors (DCPs) [13], [20], color attenuation prior [13], and haze-line prior [19], [40], [41]. Regrettably, the susceptibility to erroneous transmission estimations arising from inaccurate prior information is a prominent concern in remote sensing image dehazing. Among conventional dehazing techniques, the widely renowned approach rooted in the DCP [13] reveals a consistent observation where a solitary pixel in the RGB color channel of non-sky region exhibits an exceptionally low value. This finding enables an effective estimation of transmission map, albeit at the risk of inducing color distortion within the sky region of dehazed image. The utilization of color attenuation prior in image dehazing is introduced in [42]. By establishing a mathematical equation that incorporates disparities between image brightness and saturation, the scene depth can be inferred, thereby facilitating the acquisition of the transmission. Furthermore, [40], [41] propose a nonlocal dehazing technique, where the colors of haze-free images exhibit tight clusters within the RGB color space as the hazy images manifest haze lines, thereby the estimation of transmission map becomes feasible through the analysis of haze lines. Meanwhile, [43] introduces the concept of virtual depth, which demonstrates its utility in quantifying Earth surface coverings, enabling the estimation of transmission #Param and further facilitating effective haze removal.

B. Learning-Based Remote Sensing Image Dehazing

Although deep learning methods [9], [44], [45], [46] have demonstrated remarkable performance in various daily dehazing tasks, such as indoor and outdoor image dehazing, their direct application to remote sensing image dehazing is challenging due to distribution variations and high-resolution disparities. To this end, [28] proposes a prior-based dense attentive network that leverages dense blocks and attention blocks to restore haze-degraded remote sensing images. Subsequently, [29] designs a two-stage dehazing network, where the first stage employs an encoder–decoder architecture for coarse dehazing, while the second stage refines the coarse dehazing results. Similarly, [30] introduces a dual-step cascaded residual dense network specifically designed for single remote sensing

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image dehazing. Meanwhile, [47] proposes a deep network that combines discrete wavelet transform with a deep learning network to extract dehazed images from nonuniformly hazed remote sensing images. Furthermore, [48] enhances the temporal and spatial scopes of remote sensing image application and focuses on nonuniform haze removal. Additionally, [31] presents a multiscale architecture for removing haze degradation from remote sensing images. More recently, [32] incorporates a multimodel joint estimation module and a self-correcting module to construct a unified end-to-end dehazing network, and [2] introduces a modified normalization layer and a spatial information aggregation scheme, achieving state-of-the-art performance on both natural daily and remote sensing hazy images.

III. Proposed Method

A. Overall Pipeline

Presently, the majority of u-shaped architectures used in single image dehazing, as mentioned in works such as [2], [22], [49], [50], [51], [52], rely on intricate encoder–decoder network designs. These architectures often result in computationally demanding networks, and researchers also face challenges in designing specific number of encoders and decoders. To this end, our proposed CINet introduces a structural strategy that eliminates the need for dedicated encoders and decoders. Instead, our strategy prioritizes the management of feature flow to the reconstruction stages through the utilization of efficient skip connections. By leveraging these skip connections, we can precisely control the flow of relevant features without relying on complex encoder–decoder designs. In our pipeline, when given with a haze-degraded remote sensing image $\boldsymbol{I} \in \mathbb{R}^{3\times h\times w}$, where h and w represent the input resolution, we employ overlapping image patch embedding. This technique utilizes 3×3 convolutions [2], [34], [53] to extract high-dimensional features. The encoders and decoders exclusively employ downsampling and upsampling operations. To regulate the effective feature learning in each skip connection, we propose a MPEB. During the reconstruction phase, conventional methods typically employ concatenation and 1×1 convolution to fuse intralevel features. However, this approach has limitations in effectively aggregating information across different feature planes and fails to enable context-based fusion. In contrast, we propose an IDFM that employs lightweight adaptive transposed self-attention to fuse the features. Additionally, we incorporate a cross-level multiscale interaction module (CMIM) to facilitate information exchange between adjacent levels, which enables feature flow at different stages and mitigates information loss caused by repeated sampling operations. The restored image is generated through a 3×3 convolution and a soft residual connection [2].

B. Intralevel Dual-Dimension Fusion Module

Currently, most design [34], [53], [54], [55] based on u-shaped architectures, achieving intralevel feature fusion by utilizing channel-wise concatenation and 1×1 convolution. However, this operation demonstrates limitations in effectively aggregating information across different feature planes and fails to capture context-aware attention. An alternative approach [2] involves the utilization of a selective kernel network [56] for intralevel feature fusion. However, this operation is still accomplished through features addition and channel-wise attention, which fails to enable a more comprehensive and accurate fusion. Therefore, there is a need for an improved approach that addresses the limitations and facilitates a more robust fusion of intralevel features while considering multidimension information and promoting better information aggregation [57]. To this end, we introduce a module termed the IDFM, which utilizes features from two distinct sources to generate a self-attention map. As depicted in Fig. 3, with two given sets of input features, we initially employ the point-wise convolution to derive the query representation $\boldsymbol{Q}$ and value representation $\boldsymbol{V}$ of the features $\boldsymbol{x}_s$ from skip connections. Similarly, we acquire the key representation $\boldsymbol{K}$ and value representation $\boldsymbol{V}$ of the decoding features $\boldsymbol{x}_d$ using the same strategy. Each of these four representations are then divided into two components, corresponding to the dimensions of height and width, respectively. In height dimension, we first employ the query representation $\boldsymbol{Q}_{s,h}$ of encoding features and the key representation $\boldsymbol{K}_{d,h}$ of decoding features to compute the interactive self-attention. Subsequently, we utilize the interactive attention map of height dimension to augment both sets of height-dimension value representations. The enhanced features are combined as height-dimensional fused features. This process can be expressed as

$$\begin{aligned} \boldsymbol{A}_h &= (\boldsymbol{Q}_{s,h} \otimes \boldsymbol{K}_{d,h}) \odot \alpha_h \\ \boldsymbol{x}_h &= \boldsymbol{A}_h \otimes \boldsymbol{V}_{e,h} + \boldsymbol{A}_h^{\mathrm{T}} \otimes \boldsymbol{V}_{d,h} \end{aligned} \tag{3}$$

where $\boldsymbol{A}_h \in \mathbb{R}^{b\times w\times h\times h}$ indicates the height-dimension mutual attention map, α_h is a learnable parameter, $\boldsymbol{x}_h$ are the fused height-dimension features, and $\otimes$ denotes the matrix multiplication. Furthermore, in width dimension, we utilize the query representation $\boldsymbol{Q}_{s,w}$ of features from skip connections and the key representation $\boldsymbol{K}_{d,w}$ of decoding features to compute the interactive attention. Subsequently, we apply the interactive attention map of width dimension to enhance both two sets of

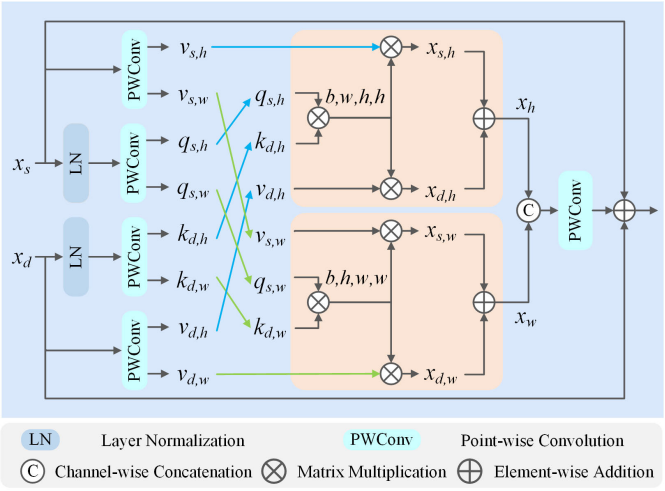


Fig. 3. Illustration of the proposed IDFM is presented. This module leverages adaptive spatial features to compute a channel-wise attention map, which is then used to perform context-based fusion.

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width-dimension value representations separately. This process can be expressed as

$$\begin{aligned} \boldsymbol{A}_w &= (\boldsymbol{Q}_{s,w} \otimes \boldsymbol{K}_{d,w}) \odot \alpha_w \\ \boldsymbol{x}_w &= \boldsymbol{A}_w \otimes \boldsymbol{V}_{e,w} + \boldsymbol{A}_w^{\mathrm{T}} \otimes \boldsymbol{V}_{d,w} \end{aligned} \tag{4}$$

where $\boldsymbol{A}_w \in \mathbb{R}^{b\times h\times w\times w}$ indicates the width-dimension mutual attention map, α_w is a learnable parameter, and $\boldsymbol{x}_w$ are the fused width-dimension features. Furthermore, we concatenate both the fused height-dimension and width-dimension features, and utilize the point-wise convolution to generate the final intralevel fused features, namely

$$\boldsymbol{x}' = \mathrm{conv}_{1\times 1}(\mathrm{concat}(\boldsymbol{x}_h, \boldsymbol{x}_w)) \tag{5}$$

where $\boldsymbol{x}'$ denotes the final intralevel fused features. This dynamic weighting mechanism enhances the model ability to adapt to various features, resulting in a context-sensitive representation of the input and further boosting the reconstruction performance. Additionally, the multidimension decoupling approach facilitates the precise integration of different features. In contrast to traditional self-attention mechanisms, our representations of $\boldsymbol{Q}$ and $\boldsymbol{K}$ are specifically designed to capture distinct global spatial features, resulting in a significant reduction in computational complexity.

C. Cross-Level Multiscale Interaction Module

Many current architectures designed for remote sensing image dehazing [2], [58], [59], [60] fail to establish effective interactions across different hierarchical levels, resulting in constrained information exchange among features at varying scales. To overcome this limitation, we propose a cross-level multiscale interaction module that promotes comprehensive global interactions. To mitigate the computational complexity arising from the squared relationship between feature resolution and spatial self-attention, we employ transposed self-attention to extract channel-wise global information. As illustrated in Fig. 2, we integrate features from two distinct levels, aligning them to a common resolution. Following this, we compute the transposed self-attention by performing matrix transposition. The value representations, which are obtained through the skip connection and upsampling process, are maintained at sizes identical to those of their corresponding input features. In our CMIM approach, the query representation $\boldsymbol{Q}$ and key representation $\boldsymbol{K}$ are generated through a distinct procedure as

$$\begin{aligned} \boldsymbol{Q} &= \mathrm{conv}_{1\times 1}(\mathrm{PU}(\mathrm{LN}(\boldsymbol{x}_l))) \\ \boldsymbol{K} &= \mathrm{conv}_{1\times 1}(\mathrm{LN}(\boldsymbol{x}_{l-1})) \end{aligned} \tag{6}$$

where PU represent the pixel-unshuffle operation, where $\boldsymbol{x}_l$ and $\boldsymbol{x}_{l-1}$ denote the feature maps at levels l and $l-1$, respectively. The attention map is subsequently computed as described in (3). Additionally, the enhanced representations of $\boldsymbol{x}_l$ and $\boldsymbol{x}_{l-1}$ can be expressed as

$$\begin{aligned} \boldsymbol{x}'_l &= \boldsymbol{x}_l + \mathrm{conv}_{1\times 1}(\boldsymbol{A} \otimes \mathrm{conv}_{1\times 1}(\boldsymbol{x}_l)) \\ \boldsymbol{x}'_{l-1} &= \boldsymbol{x}_{l-1} + \mathrm{conv}_{1\times 1}(\boldsymbol{A}^{\mathrm{T}} \otimes \mathrm{conv}_{1\times 1}(\boldsymbol{x}_{l-1})) \end{aligned} \tag{7}$$

where $\boldsymbol{x}'_l$ and $\boldsymbol{x}'_{l-1}$ denote the enhanced features after the interaction. It is important to note that the output features retain the same sizes as the input features. This characteristic enables the straightforward application of our proposed CMIM in most hierarchical frameworks. Our proposed CMIM presents two primary advantages. First, it mitigates the information loss typically induced by the repeated application of upsampling and downsampling processes. By integrating CMIM into the CINet framework, we reduce the network's susceptibility to such information degradation. Second, the multiscale features derived from one level significantly contribute to enriching the feature representations at subsequent levels, thereby enhancing the overall feature encoding.

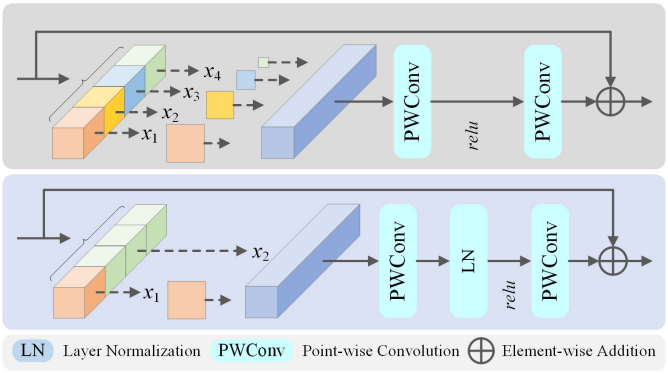


Fig. 4. Depiction of the fundamental learning module. (Top) Multiview progressive extraction module proposed in this study. (Bottom) FasterNet block incorporating one-quarter partial convolution.

D. Multiview Progressive Extraction Block

Recently, vision Transformers have emerged as a dominant approach in various image restoration tasks [2], [34], [38], [55]. However, Transformer-based methods often suffer from heavy computational complexity and longer inference times when processing the high-resolution remote sensing haze-degraded images. To address this issue, Chen et al. [36] introduce the concept of partial convolution, where one-quarter of the feature planes are transformed while keeping the remaining features constant for the subsequent layers. This approach has demonstrated superior performance compared to existing mobile networks and even some Transformer-based networks in several high-level computer vision tasks. However, due to the direct discarding of a large number of features, maintaining constant features is not the optimal approach for remote sensing feature learning. In light of the high-resolution characteristics of remote sensing hazy images, we introduce a MPEB that utilizes 1×1 convolutions for channel-wise interactions, in contrast to the traditional 3×3 convolutions employed in FasterNet [36]. As depicted in Fig. 4, the remaining three-quarters of the features are partitioned into three segments, each associated with a grouped dilated convolution possessing distinct receptive fields. Despite the differing sizes of the convolution kernels, they all share identical dilation factors. This choice of consistent dilation factors effectively mitigates the grid artifact commonly encountered when using varying dilation rates. Furthermore, based on our experimental findings, we have eliminated the original layer normalization to reduce computational overhead while maintaining

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reduced latency. The formulation of our proposed MPEB is

$$\begin{aligned} \boldsymbol{x}_i' &= \text{conv}_{(2i-1)\times(2i-1),d,g}(\boldsymbol{x}_i), \quad i = 1, 2, 3, 4 \\ \boldsymbol{x} &= \text{mlp}\big(\text{concat}\big(\boldsymbol{x}_1', \boldsymbol{x}_2', \boldsymbol{x}_3', \boldsymbol{x}_4'\big)\big) \end{aligned} \tag{8}$$

where $\text{conv}_{2i-1\times 2i-1,d,g}$ indicates the convolution layer with kernel size of $2i-1 \times 2i-1$, dilation factor of d, and group size of g. d and g conform to the follow:

$$d = \begin{cases} 1, & i = 1 \\ 3, & \text{otherwise} \end{cases}, \quad g = \begin{cases} 1, & i = 1 \\ \frac{1}{4}c, & \text{otherwise} \end{cases} \tag{9}$$

where c is the channel number of input features.

E. Soft Residual Connection and Loss Function

Furthermore, we also incorporate soft residual connection [2] to generate the restored image. Traditionally, image dehazing networks predict either the reconstructed image $\hat{J}(\boldsymbol{x})$ or the global residual $R(\boldsymbol{x}) = \hat{J}(\boldsymbol{x}) - I(\boldsymbol{x})$. However, in scenarios involving an approximation degradation model with potentially weak constraints, the use of a soft residual connection leverages the benefits of incorporating prior knowledge. This leads to a modified formulation of (1) as

$$J(\boldsymbol{x}) = K(\boldsymbol{x})I(\boldsymbol{x}) + B(\boldsymbol{x}) + I(\boldsymbol{x}) \tag{10}$$

where $K(\boldsymbol{x}) = (1/t(\boldsymbol{x})) - 1$ and $B(\boldsymbol{x}) = -((1/t(\boldsymbol{x})) - 1)\boldsymbol{A}$. In our approach, we train the network to predict $\boldsymbol{O} \in \mathbb{R}^{h\times w\times 4}$, which is subsequently decomposed into $\boldsymbol{K} \in \mathbb{R}^{h\times w\times 1}$ and $\boldsymbol{B} \in \mathbb{R}^{h\times w\times 3}$. This formulation allows the network to implicitly enforce constraints on the relationship between $K(\boldsymbol{x})$ and $B(\boldsymbol{x})$. The proposed CINet is optimized using the widely adopted $\mathcal{L}_2$ loss function, which is given by

$$\mathcal{L}_2 = \frac{1}{N}\|\boldsymbol{I}_{\text{restored}} - \boldsymbol{I}_{\text{clean}}\|_2^2 \tag{11}$$

where N is the number of pixels, and $\boldsymbol{I}_{\text{restored}}$ and $\boldsymbol{I}_{\text{clean}}$ denote the restored and clean images, respectively.

IV. Experimental Results

A. Experimental Settings

1) Implementation Specifications: Our model is trained for a total of 1000 epochs using a single NVIDIA Tesla A40 GPU, with a patch size of 256×256 and a batch size of 32. We utilize the Adam optimizer with $\beta = 0.9$ and $\beta = 0.999$, and an initial learning rate of 0.0002. The learning rate is gradually reduced to 1×10^{-6} through the use of a cosine annealing scheduler. The number of fundamental learning blocks is configured as (2, 3, 4) from the top stage to the bottom stage. During the training process, we employ data augmentation strategies, including random rotation and horizontal flipping, to improve the model's robustness. During the inference phase, we adjust its resolution to a multiple of 16×16. After the restoration process, we remove the padding from the restored image and restore it to its original size. This approach ensures that our proposed CINet is adaptable to various resolutions.

2) Datasets: We evaluate the performance of the proposed CINet on eight publicly available datasets, including StateHaze1K [61], RS-Haze [2], RICE [71], LHID [70], DHID [70], RSID [72], Dense-Haze [73], and NH-Haze [74]. StateHaze1K dataset [61] consists of three sub-datasets, StateHaze1K-thin, StateHaze1K-moderate, and StateHaze1K-thick. Each sub-dataset includes 400 pairs of synthetic remote sensing hazy images, with 320 images allocated for training, 35 for validation and 45 for testing. RS-Haze dataset [2] is a large-scale collection containing 51 300 paired images for training and 2700 paired images for testing. RICE dataset [71], which is from Google Earth, is specifically designed for remote sensing de-clouding task. It consists of 500 paired images, with 400 pairs selected for training and the remaining pairs reserved for evaluation. RSID dataset [72] contains 1000 paired images. For training our model, we randomly select 900 image pairs, while the remaining 100 pairs are employed for testing. LHID dataset [70] includes a training set and two testing sets with synthetic light haze. The training set contains 30 517 image pairs, while the two testing sets, LHID-A and LHID-B, each comprise 500 image pairs. To form a unified testing set, we combine LHID-A and LHID-B datasets, resulting in a total of 1000 testing image pairs. DHID dataset [70] is tailored for dense haze conditions. It consists of 14 490 image pairs for training and 500 image pairs for testing. Dense-Haze dataset [73] is used for dense and homogeneous haze removal, which comprises 45 image pairs for training, five pairs for validation and five pairs for testing. NH-Haze dataset [74] is focused on nonhomogeneous haze, which contains 45 image pairs for training, five pairs for validation and five pairs for testing. Additionally, we utilize a real-world remote sensing hazy dataset [37] to assess the generalization capabilities of the involved methods.

3) Evaluation Protocols: In our evaluation of synthetic hazy datasets, we utilize the following metrics, peak signal-to-noise ratio (PSNR), and structural similarity index (SSIM). The higher PSNR and SSIM values indicate superior image quality. In this work, we utilize the metrical calculating approaches in [44]. To evaluate the effectiveness of real-world image dehazing, we apply the following perceptual quality metrics, naturalness image quality evaluator (NIQE) [75], neural image assessment (NIMA) [76], multiscale image quality transformer (MUSIQ) [77], and multidimension attention network image quality assessment (ManIQA) [78]. The lower NIQE score denotes the better image quality, and vice-verse for the others. In this work, we follow the metrical calculating methods in [79] to analyze the real-world image dehazing performance.

B. Synthetic Remote Sensing Image Dehazing

To evaluate the performance of our proposed CINet on synthetic remote sensing image dehazing, we compare it with several existing dehazing methods on multiple hazy datasets, including StateHaze1K [61], RS-Haze [2], LHID [70], DHID [70], RICE [71], and RSID [72]. Table I presents the partial evaluation results, demonstrating that our CINet achieves superior performance in terms of PSNR and SSIM across both StateHaze1K and RS-Haze datasets. Specifically, our method outperforms the second-best methods by

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TABLE I

QUANTITATIVE RESULTS OF THE INVOLVED COMPARATIVE METHODS ON STATEHAZE1K [61] AND RS-HAZE [2] DATASETS. IN THIS TABLE, HIGHLIGHTED AND UNDERLINED FONT DENOTE THE HIGHEST AND SECOND-HIGHEST QUANTITATIVE PERFORMANCE, RESPECTIVELY. OUR PROPOSED CINET ACHIEVES THE BEST PERFORMANCE ACROSS ALL TESTED REMOTE SENSING HAZY DATASETS

Method	StateHaze1K-thin [61]		StateHaze1K-mod [61]		StateHaze1K-thick [61]		StateHaze1K [61]		RS-Haze [2]	
	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
DCP [13]	13.15	0.7246	9.780	0.5735	10.25	0.5850	11.06	0.6477	17.86	0.7340
NID [40]	12.87	0.6760	13.30	0.6019	14.08	0.7412	13.42	0.6730	18.81	0.7272
DehazeNet [26]	16.62	0.8313	16.01	0.8259	14.60	0.7121	15.74	0.7898	23.69	0.8071
AOD-Net [62]	15.10	0.7931	16.05	0.8354	10.77	0.5431	13.97	0.7239	22.71	0.8029
PFDN [63]	21.98	0.8913	25.54	0.9399	17.13	0.7747	21.55	0.8686	35.94	0.9480
GridDehazeNet [27]	20.09	0.8843	23.63	0.9333	15.33	0.7693	19.68	0.8623	33.99	0.9478
GCANet [64]	20.25	0.8831	22.45	0.9295	17.55	0.7911	20.08	0.8679	34.02	0.8771
PSD [65]	16.88	0.8459	19.68	0.9023	15.69	0.7538	17.42	0.8340	32.79	0.9258
DehazeFormer-T [2]	20.21	0.8856	21.54	0.9250	11.97	0.6998	17.91	0.8368	39.11	0.9587
DehazeFormer-S [2]	21.99	0.8923	22.26	0.9293	15.68	0.7746	19.98	0.8654	39.57	0.9605
DehazeFormer-B [2]	23.40	0.8985	24.53	0.9357	17.91	0.7986	21.95	0.8770	39.87	0.9616
gUNet-T [66]	21.86	0.8881	23.99	0.9307	17.65	0.7802	21.17	0.8663	37.35	0.9506
gUNet-B [66]	22.76	0.8967	25.30	0.9378	19.32	0.7950	22.46	0.8765	38.24	0.9583
gUNet-D [66]	22.25	0.8950	25.20	0.9379	18.14	0.7955	21.86	0.8761	38.63	0.9602
4KDehazing [52]	22.35	0.8939	23.73	0.9209	20.61	0.8227	22.23	0.8792	33.80	0.9351
Dehamer [67]	22.50	0.8973	25.72	0.9398	17.35	0.7957	21.86	0.8776	32.45	0.9236
C2PNet [68]	19.52	0.8758	24.40	0.9378	15.24	0.7589	19.72	0.8575	31.80	0.9159
EMPF-Net [37]	22.38	0.8947	24.42	0.9320	20.29	0.8066	22.36	0.8784	40.05	0.9610
AIDTransformer [69]	23.11	0.8942	25.06	0.9027	20.56	0.8142	22.91	0.8704	39.04	0.9604
PCSformer [46]	22.08	0.8751	23.29	0.9365	18.39	0.7727	21.25	0.8614	38.44	0.9582
RSHazeNet [39]	22.35	0.8931	24.86	0.9348	20.23	0.8222	22.48	0.8834	40.61	0.9606
PhDnet-T [44]	22.96	0.8976	26.07	0.9399	18.93	0.8032	22.65	0.8802	39.24	0.9549
PhDnet-S [44]	22.84	0.8976	26.04	0.9404	19.12	0.8064	22.67	0.8812	38.63	0.9564
PhDnet-B [44]	23.13	0.8969	26.47	0.9421	19.25	0.8069	22.95	0.8820	39.58	0.9568
CINet (ours)	**23.45**	**0.8994**	**27.12**	**0.9483**	**21.34**	**0.8340**	**23.97**	**0.8939**	**40.69**	**0.9642**

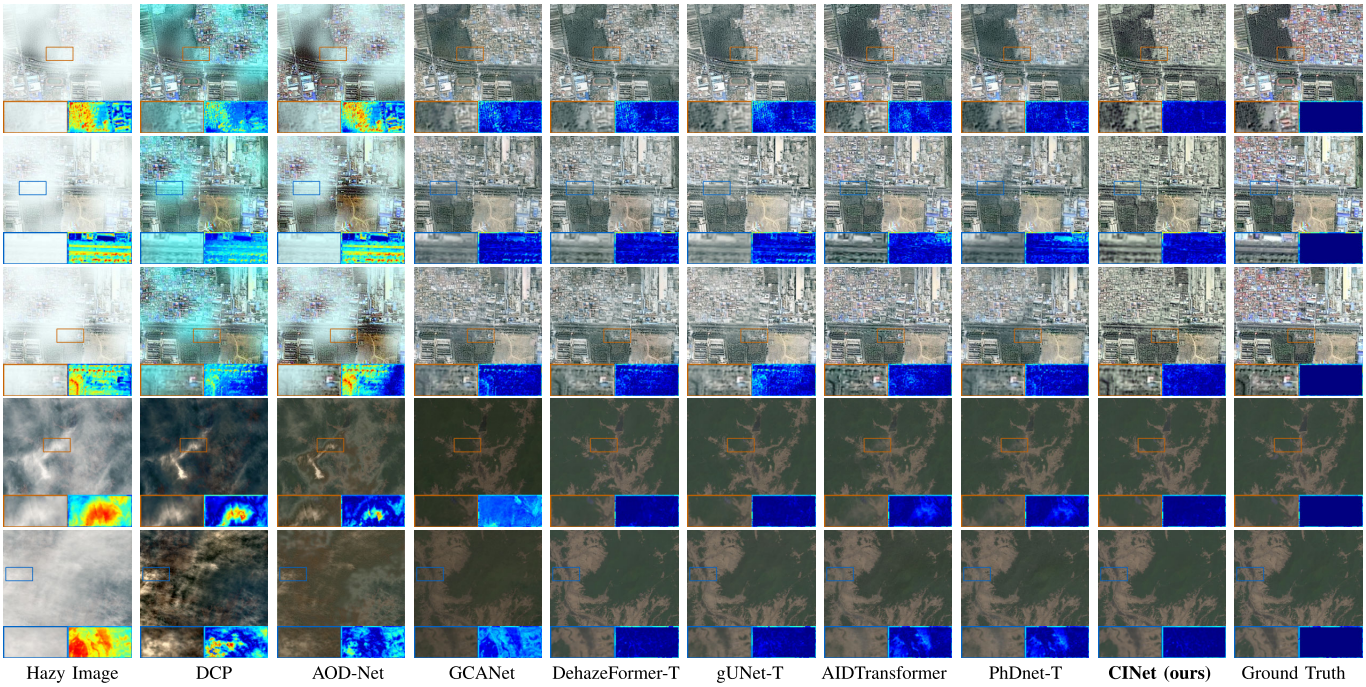


Fig. 5. Visual comparisons of the involved comparative methods and our proposed CINet on StateHaze1K [61] and RS-Haze [2] datasets. Our model achieves the best performance on haze removal and vision restoration.

0.05 dB on StateHaze1K dataset and 0.08 dB on RS-Haze dataset. Meanwhile, we provide visual comparisons between our model and other state-of-the-art dehazing methods, including DCP [13], NID [40], AOD-Net [62], GCANet [64], 4KDehazing [52], DehazeFormer-T [2], gUNet-T [66], AIDTransformer [69], PCSformer [46], and PhDnet-T [44]. Fig. 5 illustrates the dehazed images along with the corresponding error maps, where the darker content indicates the better

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TABLE II

QUANTITATIVE RESULTS OF THE INVOLVED COMPARATIVE METHODS ON LHID [70], DHID [70], RICE1 [71], RICE2 [71], AND RSID [72] DATASETS. OUR PROPOSED CINET ACHIEVES THE BEST PERFORMANCE ACROSS ALL TESTED REMOTE SENSING HAZY DATASETS

Method	LHID [70]		DHID [70]		RICE1 [71]		RICE2 [71]		RSID [72]	
	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
DCP [13]	19.17	0.7561	15.86	0.7168	16.03	0.6966	10.37	0.4491	13.33	0.7021
NID [40]	18.45	0.8187	19.11	0.7830	15.53	0.5815	13.08	0.5204	15.75	0.7816
DehazeNet [26]	21.65	0.8389	15.91	0.7252	20.94	0.8503	21.16	0.8412	17.86	0.8516
AOD-Net [62]	21.15	0.8416	15.04	0.6551	21.53	0.8550	22.48	0.8567	17.72	0.8498
PFDN [63]	26.08	0.8847	23.48	0.8604	28.26	0.9183	26.30	0.8769	21.49	0.8981
GridDehazeNet [27]	24.98	0.8710	25.01	0.8692	30.01	**0.9376**	26.20	0.8773	22.68	0.9026
GCANet [64]	25.47	0.8816	24.04	0.8611	29.30	0.8414	26.68	0.8475	21.87	0.8753
PSD [65]	24.29	0.8552	20.02	0.7240	23.22	0.8578	22.88	0.8410	18.16	0.8537
DehazeFormer-T [2]	28.58	0.8881	23.56	0.8518	27.32	0.8915	26.92	0.8718	21.36	0.8632
DehazeFormer-S [2]	29.15	0.8950	24.20	0.8600	28.05	0.9020	27.50	0.8800	22.10	0.8720
DehazeFormer-B [2]	29.73	0.8964	24.98	0.8734	28.46	0.9053	27.61	0.8826	22.84	0.8959
gUNet-T [66]	28.66	0.8811	23.07	0.8558	27.09	0.8903	26.64	0.8804	21.56	0.8720
gUNet-B [66]	29.34	0.8908	24.20	0.8617	27.53	0.9018	27.44	0.8892	22.03	0.8826
gUNet-D [66]	29.09	0.8877	23.84	0.8725	27.56	0.9029	27.98	0.8901	22.34	0.8834
4KDehazing [52]	25.73	0.8406	23.44	0.8076	27.15	0.9062	26.72	0.9140	23.01	0.9008
Dehamer [67]	26.15	0.8821	23.56	0.8473	27.34	0.9050	26.43	0.8735	21.58	0.8817
C2PNet [68]	28.27	0.8780	23.44	0.8461	26.88	0.8854	27.12	0.8795	21.59	0.8705
EMPF-Net [37]	30.18	0.8955	24.60	0.8640	29.93	0.9323	28.71	0.8952	23.09	0.9065
AIDTransformer [69]	30.21	0.9014	24.83	0.8698	30.10	0.9317	28.74	0.9029	23.35	0.9109
PCSformer [46]	28.68	0.8693	23.76	0.8375	28.00	0.8894	26.99	0.8762	19.32	0.8601
RSHazeNet [39]	30.27	0.8934	24.86	0.8880	28.09	0.9031	28.34	0.8902	22.88	0.9032
PhDnet-T [44]	29.98	0.8944	23.61	0.8492	26.89	0.9089	26.59	0.8950	22.91	0.9037
PhDnet-S [44]	30.10	0.9013	25.80	0.9062	25.96	0.9064	27.51	0.8728	23.19	0.9111
PhDnet-B [44]	30.45	0.9052	24.92	0.8964	26.88	0.9165	28.13	0.9068	23.42	0.9195
CINet (ours)	**31.04**	**0.9085**	**26.30**	**0.9088**	**30.37**	0.9369	**28.96**	**0.9189**	**23.70**	**0.9222**

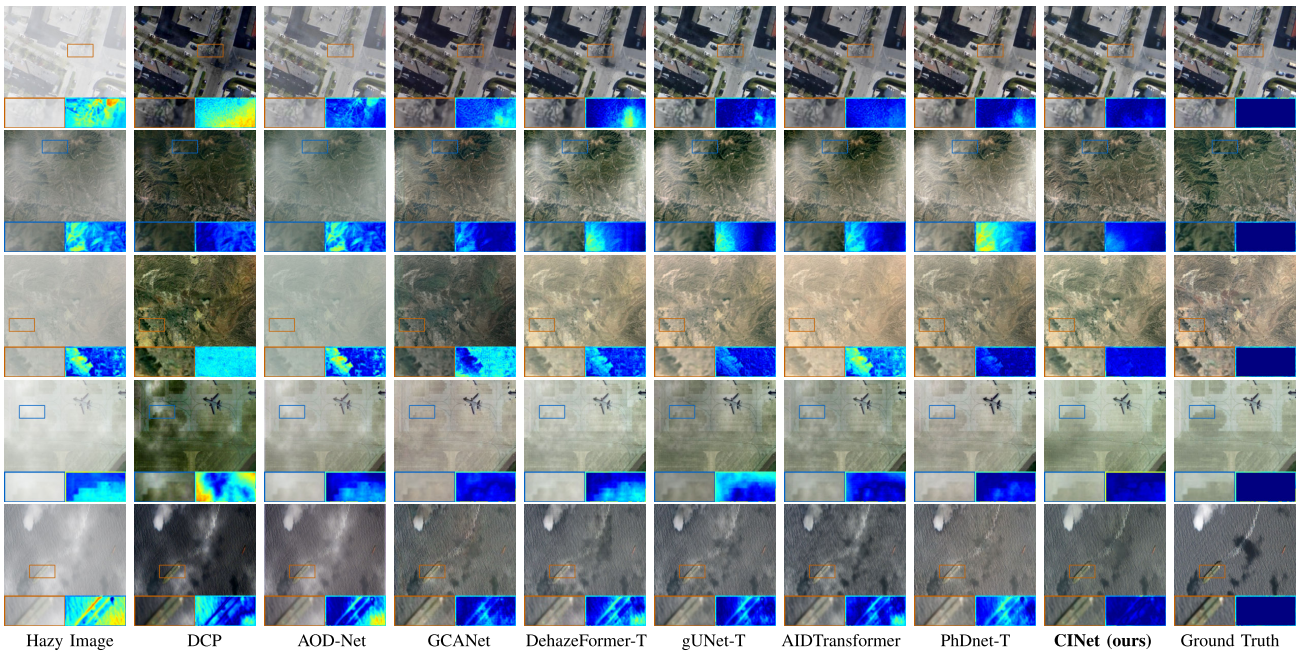


Fig. 6. Visual comparisons of the involved comparative methods and our proposed CINet on LHID [70], DHID [70], RICE [71], and RSID [72] datasets. Our model achieves the best performance on haze removal and vision restoration.

image quality. It is evident that our proposed method generates images with the minimal reconstruction error when compared to the other dehazing methods.

In addition, we also evaluate the effectiveness of our approach on LHID [70], DHID [70], RICE [71], and RSID [72] datasets. As Table II reported, our proposed model achieves performance gains of 0.59 dB on LHID dataset, 0.50 dB on DHID dataset, 0.27 dB on RICE1 dataset, 0.22 dB on RICE2 dataset, and 0.28 dB on ESID dataset over the second-best methods. Furthermore, we provide several visual results in Fig. 6, which depicts that our method effectively removes the involved several haze

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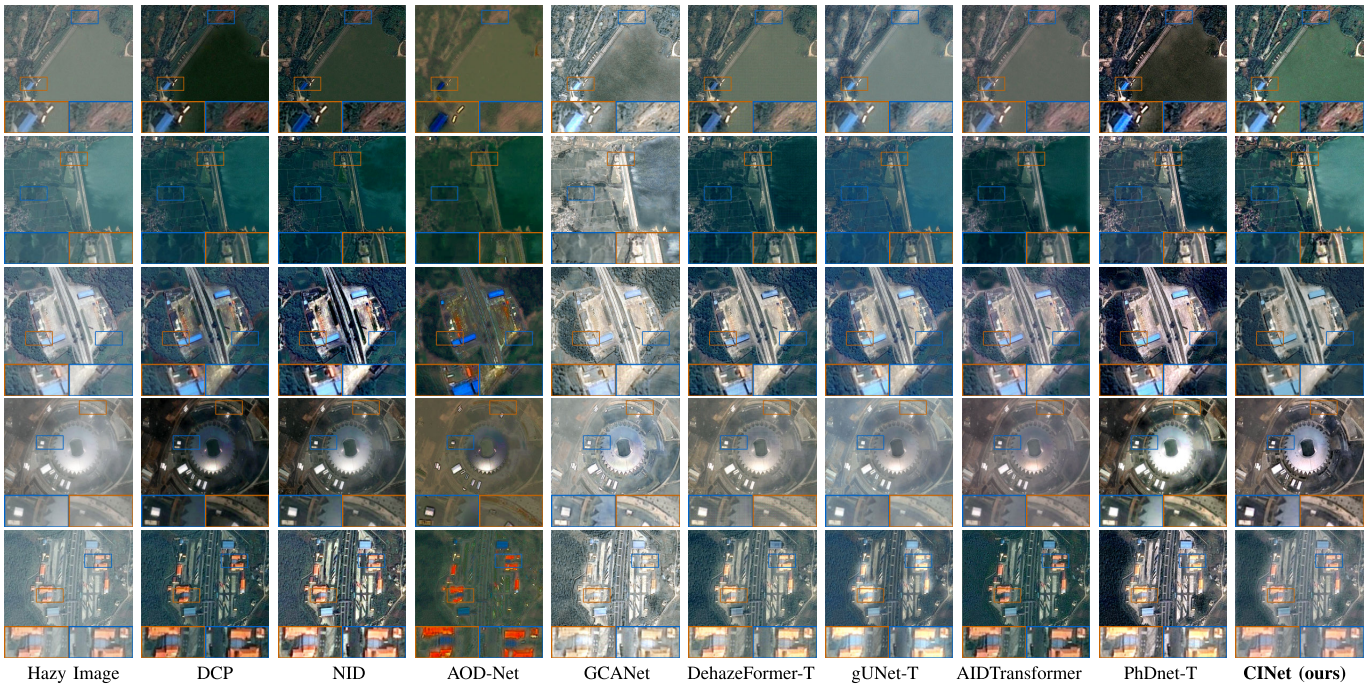


Fig. 7. Visual comparisons of the involved comparative methods and our proposed CINet on RRSD300 [37] dataset. Our model achieves the best performance on haze removal and vision restoration.

degradations while obtaining the results that are closest to the clean images.

Overall, the proposed CINet demonstrates superior quantitative and qualitative performance across the evaluated remote sensing datasets. The experimental results confirm that our model exhibits enhanced generalization capabilities across diverse datasets.

C. Real-World Remote Sensing Image Dehazing

To assess the performance of our proposed CINet for real-world remote sensing haze removal, we perform experimental comparisons using the real-world RRSD300 dataset [37]. For fair comparisons, all the reported real-world results were obtained using the respective models trained on RS-Haze dataset [2]. Table III presents the quantitative comparison of the evaluated methods, demonstrating that our model outperforms other learning-based approaches, particularly in NIQE, NIMA, and ManIQA metrics. Meanwhile, our CINet obtains the second-best performance in MUSIQ score. To further evaluate the generalizability of our CINet for real-world haze removal, we present visual comparisons in Fig. 7. These comparisons demonstrate that our method is effective to eliminate the haze degradation. Conversely, the other evaluated methods seem to bring certain damages or color distortions to the reconstructed images. As a result, through a thorough combination of quantitative and qualitative analyses, the proposed CINet outperforms other methods in real-world remote sensing image dehazing.

TABLE III

QUANTITATIVE RESULTS OF REAL-WORLD REMOTE SENSING IMAGE DEHAZING. OUR PROPOSED CINET DELIVERS THE SUPERIOR METRICAL PERFORMANCE ON THE RRSD300 DATASET

Method	NIQE	NIMA	MUSIQ	ManIQA
DCP [13]	4.069	4.928	46.33	0.2419
NID [40]	**3.989**	4.893	46.47	0.2332
DehazeNet [26]	4.464	4.355	48.81	0.2705
AOD-Net [62]	4.954	4.321	46.30	0.2538
PFDN [63]	5.071	4.588	48.32	0.2385
GridDehazeNet [27]	4.268	4.574	46.03	0.2248
GCANet [64]	4.008	4.488	**51.48**	0.2265
PSD [65]	5.292	4.223	48.14	0.2198
DehazeFormer-T [2]	8.773	4.839	42.89	0.2107
DehazeFormer-S [2]	9.310	4.596	46.47	0.2381
DehazeFormer-B [2]	8.902	4.617	45.72	0.2423
gUNet-T [66]	5.174	4.667	49.15	0.2547
gUNet-B [66]	4.280	4.629	47.95	0.2498
gUNet-D [66]	5.367	4.711	48.33	0.2531
4KDehazing [52]	4.126	4.673	47.71	0.2506
Dehamer [67]	6.186	4.496	41.51	0.2426
C2PNet [68]	4.227	4.804	48.20	0.2572
EMPF-Net [37]	3.996	<u>5.024</u>	47.93	0.2626
AIDTransformer [69]	4.202	4.881	47.21	0.2511
PCSformer [46]	9.126	4.767	44.35	0.2530
RSHazeNet [39]	4.072	4.935	48.83	<u>0.2720</u>
PhDnet-T [44]	4.197	4.822	46.65	0.2321
PhDnet-S [44]	4.277	4.749	47.58	0.2317
PhDnet-B [44]	4.009	4.958	47.60	0.2421
CINet (ours)	<u>3.992</u>	**5.122**	<u>50.38</u>	**0.2759**

D. Real-World Natural Image Dehazing

To assess the performance of our proposed CINet for real-world image dehazing, we perform experiments on Dense-Haze [73] and NH-Haze [74] datasets. As presented in Table IV, our CINet achieves superior quantitative performance across both datasets. Specifically, our CINet demonstrates performance improvements of 0.76 and 0.86 dB

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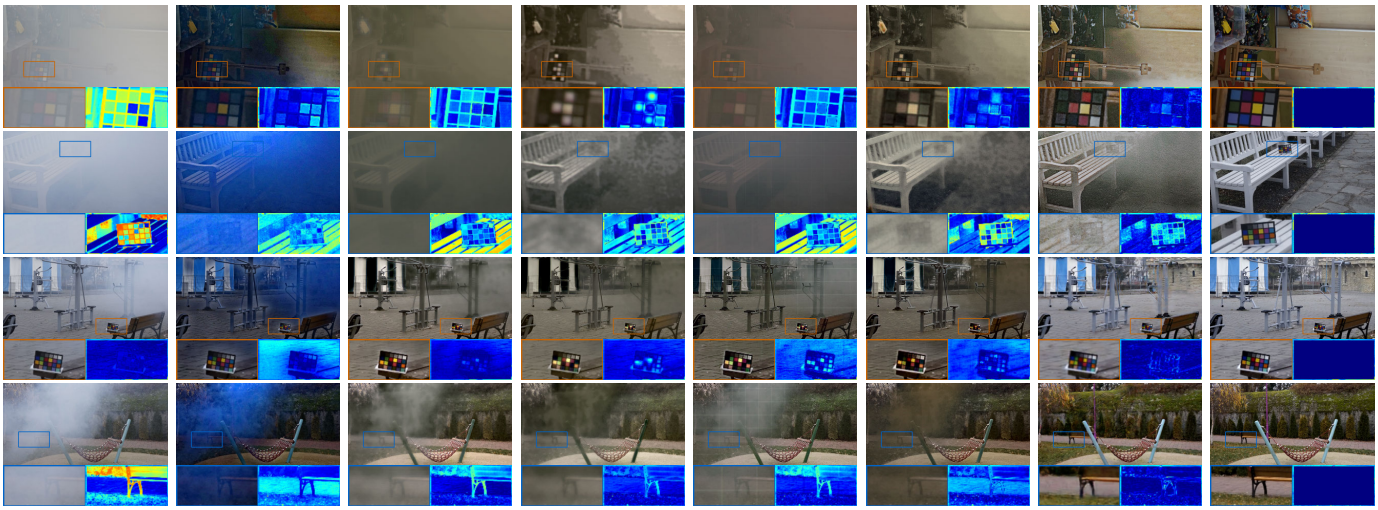


Fig. 8. Visual comparisons of the involved comparative methods and our proposed CINet on Dense-Haze [73] and NH-Haze [74] datasets. Our model achieves the best performance on haze removal and vision restoration.

TABLE IV
QUANTITATIVE DEHAZING RESULTS ON THE DENSE-HAZE [73] AND NH-HAZE [74] DATASETS. OUR PROPOSED CINET OUTPERFORMS THE OTHER EVALUATED METHODS IN BOTH PSNR AND SSIM METRICS

Method	Dense-Haze [73]		NH-Haze [74]	
	PSNR	SSIM	PSNR	SSIM
DCP [13]	10.06	0.3856	10.57	0.5196
NID [40]	12.20	0.3555	10.48	0.4442
DehazeNet [26]	13.76	0.4385	13.24	0.5401
AOD-Net [62]	11.80	0.3072	11.22	0.4193
PFDN [63]	14.84	0.3963	13.92	0.4577
GridDehazeNet [27]	11.75	0.2786	14.16	0.5000
GCANet [64]	14.16	0.4452	13.58	0.5779
PSD [65]	14.48	0.4658	13.62	0.5007
DehazeFormer-T [2]	15.02	0.4220	14.98	0.5314
DehazeFormer-S [2]	15.57	0.4663	15.41	0.5823
DehazeFormer-B [2]	16.32	0.5904	16.38	0.6288
gUNet-T [66]	14.90	0.3808	14.08	0.4857
gUNet-B [66]	15.42	0.4354	14.96	0.5301
gUNet-D [66]	15.93	0.4867	15.84	0.6038
4KDehazing [52]	14.69	0.4590	15.39	0.5529
Dehamer [67]	15.14	0.4921	14.25	0.5204
C2PNet [68]	14.76	0.4684	14.69	0.6074
EMPF-Net [37]	15.77	0.4762	15.65	0.5991
AIDTransformer [69]	15.10	0.4694	14.46	0.5511
PCSformer [46]	14.91	0.3477	14.25	0.5667
RSHazeNet [39]	15.60	0.4441	15.78	0.5302
PhDnet-T [44]	15.76	0.5657	14.66	0.5626
PhDnet-S [44]	15.23	0.4529	15.54	0.6112
PhDnet-B [44]	15.54	0.4663	15.73	0.5992
CINet (ours)	**17.08**	**0.6033**	**17.24**	**0.6495**

in PSNR on Dense-Haze and NH-Haze datasets, respectively. Furthermore, our model also outperforms the second-best DehazeFormer-B [2] in SSIM scores. In addition, we provide several visual results of the involved evaluated methods in Fig. 8. Although none of the involved methods fully eliminate the challenging haze degradation, the results generated by our approach preserve more fine-grained details and the error maps obtained by our model shows the lowest brightness. In conclusion, our proposed CINet exhibits robust performance in tackling challenging real-world dehazing tasks, surpassing the other state-of-the-art approaches in both quantitative metrics and visual comparisons. This experiment underscores the effectiveness of our proposed CINet in addressing complex real-world haze conditions, further validating the robustness and effectiveness of our proposed method.

E. Ablation Studies

To evaluate the effectiveness of our proposed several modules, we perform ablation studies in the average performance on StateHaze1K dataset [61]. These studies enable a comprehensive analysis of the contributions of each component within our proposed CINet, while also facilitating comparisons between individual components and their analogous designs.

1) Individual Components: The baseline architecture employed in our study adopts a u-shaped design, utilizing the FasterNetBlock (FNB) [36] as the foundational learning block, the 1×1 convolution for intralevel feature fusion, and the soft residual connection (SRC) [2] as output module. As shown in Table V, where w/ denotes "with," we propose the encoder-minimal and decoder-minimal framework (EDF), where the encoder stage only contains downsampling operations and the decoder stage only contains upsampling operations, resulting in a performance improvement of 0.19 dB in PSNR. Meanwhile, we observe a reduction in both the number of #Param and the floating-point operations (FLOPs) required for processing an image of size $3 \times 256 \times 256$, which indicates that our architecture design improves the model performance while reducing the computational burden. Furthermore, we replace the 1×1 convolution with our proposed IDFM for intralevel feature fusion, leading to a performance gain of 0.44 dB. Subsequently, we integrate our proposed CMIM into the model, which achieves a performance gain of 0.38 dB. In addition, we employ our proposed MPEB to be the basic learning block, which advances the

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TABLE V

ABLATION EXPERIMENTS OF THE INDIVIDUAL COMPONENTS. EACH PROPOSED COMPONENT CONTRIBUTES SIGNIFICANTLY AND POSITIVELY INFLUENCES THE OVERALL PERFORMANCE OF OUR PROPOSED MODEL IN REMOTE SENSING IMAGE DEHAZING

Model	PSNR	SSIM	#Param	FLOPs
Baseline	21.91	0.8644	628.0 K	1.922 G
w/ EDF	22.10	0.8672	**610.3 K**	**1.782 G**
w/ IDFM	22.54	0.8736	639.7 K	2.015 G
w/ CMIM	22.92	0.8860	652.4 K	2.647 G
w/ MPEB	**23.97**	**0.8939**	646.2 K	2.624 G

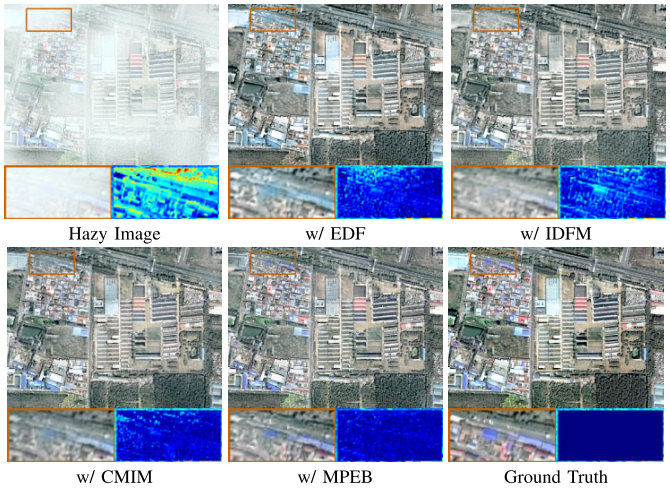


Fig. 9. Visual comparisons of the ablation studies. Each proposed component contributes significantly and positively influences the overall performance of our proposed model in remote sensing image dehazing.

simple FNB by 1.05 dB with marginally decreased calculating complexity. Therefore, each individual component contributes positively to the overall quantitative performance of our model in remote sensing image dehazing. Furthermore, we also visualize several samples in Fig. 9, which demonstrates that each component presents positive effect to the overall visual performance.

2) Intralevel Features Fusion: To further evaluate the performance of our proposed intralevel fusion module, we conduct comparative experiments of our proposed IDFM against commonly used methods, including 1×1 convolution, selective kernel fusion (SKFusion) [2], and physics-aware intralevel fusion module (PIFM) [37]. As Table VI shown, our module demonstrates superiority over the traditional convolution, with a notable improvement of 0.95 dB in PSNR. Additionally, when compared to SKFusion and PIFM, our IDFM outperforms both by 0.83 and 0.52 dB in PSNR, respectively. Meanwhile, all four intralevel feature fusion modules show similar computational burdens. The results highlighting the effectiveness of IDFM in enhancing hazing performance.

3) Basic Learning Block: To evaluate the effectiveness of our proposed MEPB, we conduct experiments to compare its performance with FNB [36] and our proposed MPEB without the multiscale design. The results presented in Table VII demonstrate the superior performance of our MEPB in remote sensing image dehazing. Specifically, our MEPB without multiscale design achieves a PSNR improvement of 1.05 dB over FNB. Meanwhile, with the multiscale design, our MPEB obtains a performance gain of 0.89 dB. Additionally, this performance enhancement is attained with a reduced number of #Param and FLOPs, which highlights the effectiveness of our MEPB, reinforcing its significance as a substantial advancement for remote sensing image dehazing.

TABLE VI

ABLATION EXPERIMENTS OF THE INTRALEVEL FEATURE FUSION MODULE. OUR PROPOSED IDFM OUTPERFORMS THE OTHER ALTERNATIVE DESIGNS

Component	PSNR	SSIM	#Param	FLOPs
Convolution	23.02	0.8826	620.9 K	2.583 G
SKFusion	23.14	0.8843	**612.6 K**	**2.218 G**
PIFM	23.45	0.8880	615.3 K	2.297 G
IDFM	**23.97**	**0.8939**	646.2 K	2.624 G

TABLE VII

ABLATION EXPERIMENTS OF THE FUNDAMENTAL LEARNING BLOCK. OUR PROPOSED MPEB ACHIEVES SUPERIOR PERFORMANCE, WHILE CONCURRENTLY REDUCING BOTH THE PARAMETER COUNT AND COMPUTATIONAL COMPLEXITY

Component	PSNR	SSIM	#Param	FLOPs
FNB	22.92	0.8860	652.4 K	2.647 G
MPEB (w/o multi-scale)	23.08	0.8894	660.5 K	2.773 G
MPEB (w/ multi-scale)	**23.97**	**0.8939**	**646.2 K**	**2.624 G**

F. Efficiency Analysis

In this experiment, we utilize three metrical scores to conduct efficiency analysis of different methods, include the number of #Param, FLOPs required for processing an image of size $3 \times 256 \times 256$, and the average inference time when processing 100 images. As Table VIII depicted, DehazeNet and AOD-Net achieves the best efficiency performance. In addition, compared to the other comparative algorithms, our method achieves the second-best performance in parameter quantity and the best performance in computational burden. In inference time, our model achieves the best performance compared to recently well-performing methods. Furthermore, we visualize this table in Fig. 1, which demonstrates that our model obtains the best trade-off between the model efficiency and dehazing performance.

V. LIMITATIONS

As depicted in Fig. 8, our method demonstrates superior capability in generating dehazed images with enhanced details and improved background recognition when dealing with complex natural haze degradations [73], [74]. However, our method does not entirely eliminate the degradation, leading to suboptimal quantitative and qualitative outcomes. The other limitation of our approach is the variability in characteristics among different datasets used during training. This necessitates training our methods separately, resulting in limited generalization capabilities in real-world remote sensing haze removal. Therefore, our approach not fully addresses the requirements of diverse and complex distributed haze scenarios.

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TABLE VIII
EFFICIENCY ANALYSIS OF THE INVOLVED COMPARATIVE METHODS. OUR PROPOSED CINET STRIKES THE OPTIMAL BALANCE BETWEEN THE MODEL EFFICIENCY AND PERFORMANCE METRICS

Method	#Param	FLOPs	Inference time
DehazeNet [26]	7.777 K	505.4 M	0.0106 s
AOD-Net [62]	**1.761 K**	**114.4 M**	**0.0004 s**
PFDN [63]	11.57 M	38.27 G	0.0885 s
GridDehazeNet [27]	955.7 K	21.43 G	0.0161 s
GCANet [64]	702.5 K	18.57 G	0.0065 s
PSD [65]	5.316 M	288.5 G	0.0593 s
DehazeFormer-T [2]	686.4 K	6.251 G	0.0202 s
DehazeFormer-S [2]	1.283 M	12.06 G	0.0400 s
DehazeFormer-B [2]	2.514 M	23.63 G	0.0937 s
gUNet-T [66]	844.9 K	2.808 G	0.0125 s
gUNet-B [66]	2.741 M	9.270 G	0.0374 s
gUNet-D [66]	5.270 M	17.89 G	0.0674 s
4KDehazing [52]	34.55 M	103.9 G	0.0518 s
Dehamer [67]	29.44 M	59.28 G	0.0406 s
C2PNet [68]	7.169 M	461.0 G	0.1104 s
EMPF-Net [37]	300.2 K	2.897 G	0.0290 s
AIDTransformer [69]	10.86 M	197.9 G	0.4423 s
PCSformer [46]	3.675 M	27.41 G	0.1740 s
RSHazeNet [39]	1.190 M	10.04 G	0.0221 s
PhDnet-T [44]	2.954 M	9.067 G	0.0339 s
PhDnet-S [44]	5.271 M	17.06 G	0.0858 s
PhDnet-B [44]	9.905 M	33.03 G	0.0935 s
CINet (ours)	646.2 K	2.624 G	0.0153 s

VI. CONCLUSION

In this study, we introduce cross-level interaction and intralevel fusion network, a notable advancement in remote sensing image dehazing. By employing an encoder-minimal and decoder-minimal architecture, alongside incorporating modules such as IDFM, cross-level multiview interaction module and MPEB, our model effectively balances image quality and computational efficiency. Extensive experimental evaluations across various haze-degraded datasets reveal that our model significantly surpasses existing state-of-the-art methods in remote sensing image dehazing. In the future, our research will focus on exploring the discrepancies among various datasets and constructing a unified remote sensing hazy dataset that encompasses a wide range of situations.

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